

Intro Typing and Simple Circuits

Piter Garcia

March 20, 2026

Packet Use

This printable lesson packet is generated from the paired Markdown guide. The Markdown version remains the clickable, neurodivergent-friendly working copy; this PDF carries the same lesson substance in a formal handout format. Evidence claims in this packet are based on transcript-derived lesson notes and the cleaned transcript files in this workspace.

Quick Scan

- **Grade:** Kindergarten
- **Grade Hub:** Notes [../kindergarten-Notes.md]
- **Schedule Reference:** Spring2026_TeachingSchedule_Derek.md [../school/Spring2026_TeachingSchedule_Ignite.md]
- **Workbook Sheet:** K in Teaching_Placement-IGNITE Curriculum_ GCSD25-26.xlsx
- **Lesson Focus:** Short keyboarding routines paired with a teacher-guided introduction to simple circuits using Play-Doh and classroom-safe materials.
- **CS Alignment Strength:** Foundational CS + physical computing bridge
- **LaTeX Source:** intro-typing-and-simple-circuits.tex [./intro-typing-and-simple-circuits.tex]
- **Bib File:** intro-typing-and-simple-circuits.bib [./intro-typing-and-simple-circuits.bib]
- **Artifacts Folder:** artifacts/ [./artifacts/]

Lesson Identity

- **Likely Class Blocks:** Day 1 – 12:25-1:15 – Justinger, Day 2 – 12:25-1:15 – Kesys, Day 4 – 12:25-1:15 – Pum
- **Main Lesson Family:** Intro Typing and Simple Circuits
- **Primary Evidence Base:** Transcript-derived notes plus source transcripts from this teaching-placement workspace.
- **Primary External Source Type:** Official curriculum/provider materials.

What We Are Teaching

- home-row style finger placement and space-bar recognition
- following a step sequence before receiving the hands-on science task
- identifying battery, bulb, and connection points in a circuit activity
- understanding that a working system depends on complete connections

CS Standards Alignment

- **1A-CS-02:** Use appropriate terminology to describe common physical components of computing systems. Why it fits here: Students name and use keyboard parts and circuit components such as battery and bulb.
- **1A-AP-08:** Model daily processes by creating and following algorithms. Why it fits here: Both the typing warm-up and the circuit build depend on following steps in order.
- **1A-AP-11:** Decompose the steps needed to solve a problem into a precise sequence of instructions. Why it fits here: Teacher instruction explicitly breaks the task into one step at a time before students try independently.

NYS / Local Curriculum Alignment

- Aligned to the local IGNITE kindergarten workbook sequence through keyboard entry routines, safe material handling, and early engineering-style problem solving.
- Supports New York-facing expectations for digital fluency and hands-on scientific observation by connecting device use with simple cause-and-effect investigation.
- Bridges computing systems with early physical computing by showing that both keyboard routines and circuits depend on correct inputs and complete connections.

Lesson Content and Concepts

- sequence
- input
- system parts
- closed circuit
- debugging by checking connections

Evidence / Proof of Instruction

- **Transcript-derived note:** 260305 Kindergarten Justinger Typing Circuits [../transcript-derived-lessons/260305-kindergarten-justinger-typing-circuits.md] The lesson note shows a two-part structure: typing first, then Play-Doh circuits after students complete the warm-up.
- **Transcript-derived note:** 260305 Kindergarten Justinger Circuits Summary [../transcript-derived-lessons/260305-kindergarten-justinger-circuits-summary.md] The planning summary confirms that the circuit activity was intentionally paired with the typing session, not improvised.
- **Source transcript:** 260305 Kindergarten Justinger Typing Circuits [../transcripts/260305-kindergarten-justinger-typing-circuits.md] The transcript records direct modeling of F, J, and space, followed by explicit directions to split Play-Doh, avoid mixing colors, and build with provided materials.
- **Likely student proof:** Students completing the typing prompt, assembling a working or near-working circuit, and verbally naming parts count as direct evidence of instruction.

Assessment Evidence

- Student finds required keys with prompting reduced over time.
- Student can identify at least one circuit part and explain that contact matters.
- Student participates safely in the material routine and attempts the build sequence.

UDL and Inclusion Supports

Multiple Representation

- Show keys visually and physically point to them on the keyboard.
- Model circuit parts one at a time before handing out all materials.
- Use color and spatial language consistently: here, there, together, connected.

Multiple Action / Expression

- Allow partner-supported keyboarding and hand-over-hand support when needed.
- Offer pre-portioned materials and physically larger examples.
- Accept explanation through pointing, showing, or brief oral response.

Multiple Engagement

- Use the science activity as a motivating payoff for the typing warm-up.
- Keep the typing segment short and success-based.
- Make the transition clear so students know when one task ends and the next begins.

How We Are Already Making It Inclusive

- The transcript shows strong teacher modeling, immediate feedback, and a high-support release into practice.
- Students were given a clear transition reward structure, which helped sustain attention through the warm-up.
- The hands-on circuit task created a non-text entry point for students who struggle with abstract explanation.

How to Improve Inclusion Next Time

- Add visual step mats for both the keyboard task and the circuit build.
- Prepare one complete exemplar circuit at each table for students who need a concrete anchor.
- Add sentence stems such as “It worked because...” and “We need to fix...”
- Create a quieter typing option or reduced-key version for students who fatigue quickly.

Materials and Setup

- Student devices
- Typing activity or typing game
- Play-Doh
- Battery pack or coin battery demo
- Bulb/LED and connectors
- Projected model

Teacher Moves / Script / Routines

- Model the typing task with only the needed keys visible in student attention.
- Pause before the circuit release to restate one safety rule and one build rule.
- Circulate first to tables that need connection help rather than re-explaining to the whole group.
- Close by asking students to show the part that made the system work.

Common Errors and Debugging

- Students may confuse keyboard keys or lose left/right hand positioning.
- Students may mix Play-Doh colors before building and lose structural clarity.
- Circuit failures will often come from incomplete contact or reversed/loose placement rather than misunderstanding the concept.

Extensions and Simplifications

- Extension: have students compare two circuit builds and say which connection is stronger.
- Simplification: use a teacher demo circuit if student independence is too high a load.
- Extension: connect typing “input” to the idea that circuits also need the right path to work.

Related Local Materials

- No direct local lesson packet linked yet.

Artifacts Checklist

- Compiled lesson PDF in **artifacts/**
- At least one screenshot, student example, or setup photo
- One short assessment or observation record
- Updated standards/source bibliography once the **.bib** file is revised
- Any teacher reflection or revision notes after reteaching

Selected Official Sources

Selected official sources that informed this packet are listed below. ocite*

References